Hypothesis Testing

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: The logic of hypothesis testing**

State the null and alternative hypotheses for testing whether a new fertilizer increases the mean crop yield μ, given a historical mean $\mu_0 = 50$ kg. Classify the test as one-tailed or two-tailed.

$H_0 : \mu = 50$. $H_a : \mu > 50$. This is a one-tailed (right-tailed) test, since the alternative specifies a direction.

2 **[Foundational] • Target: The logic of hypothesis testing**

Explain the difference between a one-tailed and a two-tailed hypothesis test, giving an example scenario for each.

A one-tailed test is used when the alternative hypothesis specifies a direction (e.g. testing whether a new drug \emph{increases} recovery rate). A two-tailed test is used when the alternative simply states the parameter differs from the claimed value, without specifying direction (e.g. testing whether a machine's mean output has changed at all, up or down).

3 **[Foundational] • Target: The logic of hypothesis testing**

A test has $H_0 : \mu = 75$ vs $H_a : \mu \neq 75$, and the sample data leads to rejecting H_0 at the 5% significance level. State, without further calculation, whether this data would also lead to rejection at the 1% level, justifying your answer.

Not necessarily — a stricter significance level (1%) requires stronger evidence (a smaller p-value) to reject H_0. If the p-value is very small (well below 0.01), rejection would still occur at 1%; but if the p-value lies between 0.01 and 0.05, H_0 would be rejected at 5% but not at 1%.

4 **[Foundational] • Target: Hypothesis test for a population mean (known σ)**

A machine fills bags with a claimed mean of 500g, with known population standard deviation $\sigma = 15$g. A sample of $n = 36$ bags has mean $\bar{x} = 494$g. Test at the 5% level whether the mean is less than 500g, using $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ and critical value -1.645.

$z = \frac{494 - 500}{15/\sqrt{36}} = \frac{-6}{2.5} = -2.4$. Since $-2.4 < -1.645$, we reject H_0. There is significant evidence at the 5% level that the mean fill weight is less than 500g.

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5 [Foundational] • Target: Hypothesis test for a population mean (known σ)

A population has known standard deviation $\sigma = 8$. A sample of $n = 64$ has mean $\bar{x} = 103$, against a claimed mean $\mu_0 = 100$. Test at the 1% level whether the mean exceeds 100, using critical value $z^* = 2.326$.

$z = \frac{103 - 100}{8/\sqrt{64}} = \frac{3}{1} = 3$. Since $3 > 2.326$, we reject H_0. There is significant evidence at the 1% level that the mean exceeds 100.

6 [Foundational] • Target: Two-tailed tests for a population mean

A standardized test historically has mean $\mu_0 = 100$ with $\sigma = 15$. A sample of $n = 25$ students taking a new prep course has mean $\bar{x} = 107$. Test at the 5% level (two-tailed) whether the course changes the mean score, using critical values ± 1.96.

$z = \frac{107 - 100}{15/\sqrt{25}} = \frac{7}{3} \approx 2.333$. Since $|2.333| > 1.96$, we reject H_0. There is significant evidence at the 5% level that the prep course changes the mean score.

7 [Foundational] • Target: Two-tailed tests for a population mean

Using $z \approx 2.33$ from the previous problem, find the two-tailed p-value, given $\Phi(2.33) \approx 0.9901$, and confirm the conclusion.

$p-\text{value} = 2(1 - \Phi(2.33)) = 2(1 - 0.9901) = 2(0.0099) = 0.0198$. Since $0.0198 < 0.05$, we reject H_0, confirming the earlier conclusion.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Two-tailed tests for a population mean

A sample of $n = 50$ has mean $\bar{x} = 48.2$, against a claimed mean $\mu_0 = 50$ with $\sigma = 6$. Test at the 5% level (two-tailed) whether the mean differs from 50, using critical values ± 1.96.

$z = \frac{48.2 - 50}{6/\sqrt{50}} = \frac{-1.8}{0.8485} \approx -2.121$. Since $|-2.121| > 1.96$, we reject H_0. There is significant evidence at the 5% level that the mean differs from 50.

9 [Intermediate] • Target: Hypothesis test for a Poisson mean

A website historically crashes at a rate of $\lambda_0 = 4$ per month. After a software update, 1 crash is observed in a randomly selected month. Test at the 5% level whether the crash rate has decreased, using $P(X \le 1) \approx 0.0916$ for $X \sim \text{Po}(4)$.

$H_0 : \lambda = 4$. $H_a : \lambda < 4$. $p-\text{value} = P(X \le 1) = 0.0916$. Since $0.0916 > 0.05$, we do not reject H_0. There is insufficient evidence at the 5% level that the crash rate has decreased.

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10 [Intermediate] • Target: Hypothesis test for a Poisson mean

Explain why an exact Poisson probability, rather than a normal approximation, is used to test a hypothesis about a Poisson mean when λ_0 is small.

When λ_0 is small, the Poisson distribution is noticeably skewed and a normal approximation would be unreliable, so exact Poisson probabilities give a more accurate p-value.

11 [Intermediate] • Target: Further Poisson mean tests

A call center historically receives calls at $\lambda_0 = 6$ per hour. During a promotional event, 12 calls are received in one hour. Test at the 5% level whether the call rate has increased, using $P(X \geq 12) \approx 0.0201$ for $X \sim \text{Po}(6)$.

$H_0 : \lambda = 6$. $H_a : \lambda > 6$. $p\text{-value} = P(X \geq 12) = 0.0201$. Since $0.0201 < 0.05$, we reject H_0. There is significant evidence at the 5% level that the call rate has increased during the promotional event.

12 [Intermediate] • Target: Further Poisson mean tests

A factory's defect rate is historically $\lambda_0 = 3$ per shift. During a trial of new equipment, 8 defects are observed in one shift. Test at the 5% level whether the defect rate has increased, using $P(X \geq 8) \approx 0.0119$ for $X \sim \text{Po}(3)$.

$H_0 : \lambda = 3$. $H_a : \lambda > 3$. $p\text{-value} = P(X \geq 8) = 0.0119$. Since $0.0119 < 0.05$, we reject H_0. There is significant evidence at the 5% level that the defect rate has increased.

13 [Intermediate] • Target: Hypothesis test for a binomial proportion

A company claims 40% of its customers use its mobile app ($p_0 = 0.4$). In a sample of $n = 15$ customers, 10 report using the app. Test at the 5% level whether the true proportion is greater than 40%, using $P(X \geq 10) \approx 0.0338$ for $X \sim B(15, 0.4)$.

$H_0 : p = 0.4$. $H_a : p > 0.4$. $p\text{-value} = P(X \geq 10) = 0.0338$. Since $0.0338 < 0.05$, we reject H_0. There is significant evidence at the 5% level that more than 40% of customers use the app.

14 [Intermediate] • Target: Hypothesis test for a binomial proportion

Explain why the binomial hypothesis test uses the exact binomial distribution rather than a normal approximation when n is small.

For small n, the binomial distribution can be quite skewed (especially if p_0 is far from 0.5), so a normal approximation would be unreliable; exact binomial probabilities give a more accurate p-value.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 2.01 Exponentials & Logarithms

A coin is tested for fairness ($p_0 = 0.5$). In 20 tosses, 6 heads are observed. Test at the 5% level (two-tailed) whether the coin is biased, using $P(X \leq 6) \approx 0.0577$ for $X \sim B(20, 0.5)$ (doubled for a two-tailed test).

$H_0 : p = 0.5$. $H_a : p \neq 0.5$. Two-tailed $p\text{−value} = 2 \times P(X \leq 6) = 2(0.0577) = 0.1154$. Since $0.1154 > 0.05$, we do not reject H_0. There is insufficient evidence at the 5% level that the coin is biased.

16 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 3.01 Advanced Trigonometry

A politician claims 60% of voters support a new policy ($p_0 = 0.6$). In a sample of 12 voters, only 3 support it. Test at the 5% level whether support is less than 60%, using $P(X \leq 3) \approx 0.0153$ for $X \sim B(12, 0.6)$.

$H_0 : p = 0.6$. $H_a : p < 0.6$. $p\text{−value} = P(X \leq 3) = 0.0153$. Since $0.0153 < 0.05$, we reject H_0. There is significant evidence at the 5% level that support is less than 60%.

17 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 4.01 Differentiation

In the website crash-rate test, describe what a Type I error would mean in context, and what a Type II error would mean.

A Type I error would mean concluding the crash rate has decreased when it actually hasn't (rejecting a true H_0) — wrongly crediting the software update. A Type II error would mean failing to detect a real decrease in the crash rate (failing to reject a false H_0) — missing genuine improvement from the update.

18 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 5.01 Integration

Explain what is meant by the significance level of a hypothesis test, and state the probability of a Type I error in a test conducted at the 5% significance level.

The significance level α is the probability of rejecting a true null hypothesis (a Type I error), chosen in advance by the person conducting the test. At the 5% significance level, $\alpha = 0.05$, so there is a 5% probability of a Type I error.

19 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 6.01 Sequences & Series

A researcher conducts a test at $\alpha = 0.10$ instead of the more common $\alpha = 0.05$. Explain how this affects the probability of Type I and Type II errors, compared to using $\alpha = 0.05$.

Using a larger $\alpha = 0.10$ increases the probability of a Type I error (the rejection region is larger, so a true H_0 is more likely to be rejected by chance) but decreases the probability of a Type II error (it becomes easier to reject H_0, so a false H_0 is more likely to be correctly rejected) — there is a direct tradeoff between the two error types as α changes.

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20 [Challenging] • Target: A well-designed word problem: testing a new manufacturing process • Cross-Lesson: 7.01 Vectors

This question has four parts. A factory's current process produces components with mean breaking strength $\mu_0 = 250$ N and known $\sigma = 20$ N. A new process gives a sample of $n = 40$ components with mean $\bar{x} = 257$ N. (a) State H_0 and H_a for testing whether the new process increases mean breaking strength (one-tailed). (b) Compute the test statistic z. (c) Using critical value $z^* = 1.645$ (5% level), state the conclusion. (d) Quality control also monitors defects; the historical rate is $\lambda_0 = 5$ per batch, but a batch from the new process has only 1 defect. Using $P(X \le 1) \approx 0.0404$ for $X \sim \text{Po}(5)$, test at the 5% level whether the new process has reduced the defect rate.

(a) $H_0 : \mu = 250$. $H_a : \mu > 250$. (b) $z = \frac{257 - 250}{20/\sqrt{40}} = \frac{7}{3.1623} \approx 2.214$. (c) Since $2.214 > 1.645$, we reject H_0 : significant evidence at the 5% level that the new process increases mean breaking strength. (d) $H_0 : \lambda = 5$, $H_a : \lambda < 5$. $p\text{-value} = P(X \le 1) = 0.0404 < 0.05$, so we reject H_0: significant evidence at the 5% level that the new process has reduced the defect rate.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The logic of hypothesis testing

For a hypothesis test at the 5% (two-tail) significance level, find the critical z-value.

a. 2.576

b. 1.645

c. **1.96** ✓

d. 1.282

The critical value for a 5% (two-tail) test is $z = 1.96$.

22 [Foundational] • Target: Hypothesis test for a population mean (known σ)

A hypothesis test at the 1% significance level gives a p-value of 6%. What is the decision?

a. Test is inconclusive

b. Reject H_0

c. Increase the sample size

d. **Do not reject** H_0 ✓

Since the p-value $\ge \alpha$ ($6\% \ge 1\%$), we do not reject H_0.

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23 [Foundational] • Target: Two-tailed tests for a population mean

For a hypothesis test at the 5% (one-tail) significance level, find the critical z-value.

a. 1.96

b. **1.645** ✓

c. 1.282

d. 2.576

The critical value for a 5% (one-tail) test is $z = 1.645$.

24 [Foundational] • Target: Hypothesis test for a Poisson mean

A test of $H_0 : \mu = 47$ uses a sample of size $n = 9$, with $\sigma = 8$ (known) and $\bar{x} = 55$. Find the test statistic $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$.

a. **3** ✓

b. -3

c. 4

d. 9

$\sigma/\sqrt{n} = 8/3 = \frac{8}{3}$. $\bar{x} - \mu_0 = 8$. $z = 8 \div \frac{8}{3} = 3$.

25 [Foundational] • Target: Further Poisson mean tests

A two-tail hypothesis test has critical value $z = \pm 2.326$ and test statistic $z = 2$. What is the decision?

a. Accept H_1 unconditionally

b. Insufficient information

c. Reject H_0

d. **Do not reject** H_0 ✓

Since $|2| \leq 2.326$, we do not reject H_0.

26 [Foundational] • Target: Hypothesis test for a binomial proportion

A one-tail (lower) hypothesis test has critical value $z = -2.326$ and test statistic $z = -1$. What is the decision?

a. **Do not reject** H_0 ✓

b. Reject H_0

c. Insufficient information

d. Accept H_1 unconditionally

Since $-1 \geq -2.326$, we do not reject H_0.

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27 [Foundational] • Target: Further binomial proportion tests

A sample of size $n = 49$ has mean $\bar{x} = 44$, drawn from a population with $\mu_0 = 47$ and known $\sigma = 10$. Find the test statistic z, correct to 2 decimal places.

a. **-2.10 ✓**

b. -3.60

c. -1.60

d. -1.10

Working through this statistics step: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{44 - 47}{10/\sqrt{49}} \approx -2.10$

28 [Foundational] • Target: Type I and Type II errors, and significance level

A population is modeled by $P(t) = P_0 \cdot 2^{t/10}$ with $P_0 = 200$. Find t when $P(t) = 400$.

a. 20

b. **10 ✓**

c. 11

d. 1

$2^{t/10} = 400/200 = 2 \Rightarrow t/10 = 1 \Rightarrow t = 10$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: testing a new manufacturing process

Two lines have direction vectors $\mathbf{d_1} = (-2, 3, -5)$ and $\mathbf{d_2} = (4, -6, 10)$. What is the relationship between the two lines (assuming they are coplanar)?

a. neither parallel nor perpendicular

b. perpendicular

c. identical

d. **parallel ✓**

$\mathbf{d_2} = -2\mathbf{d_1}$, so the lines are parallel.

30 [Intermediate] • Target: The logic of hypothesis testing

A population is modeled by $P(t) = P_0 \cdot 2^{t/5}$ with $P_0 = 100$. Find $P(5)$.

a. 100

b. 102

c. 300

d. **200 ✓**

$P(5) = 100 \cdot 2^{5/5} = 100 \cdot 2^1 = 200$.

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31 [Intermediate] • Target: Hypothesis test for a population mean (known σ)

A population grows according to $\frac{dP}{dt} = kP$. If the population triples every 1 year, find k.

a. $\log(4)$

b. $1 + \log(3)$

c. $\log(3)$ ✓

d. $\log(3) + 2$

$e^{kT} = 3 \Rightarrow k = \frac{\ln 3}{1}$.

32 [Intermediate] • Target: Two-tailed tests for a population mean

A population grows according to $\frac{dP}{dt} = kP$. If the population triples every 2 years, find k.

a. $\frac{\log(4)}{2}$

b. $2\log(3)$

c. $\frac{\log(3)}{2}$ ✓

d. $\frac{\log(3)}{2} + 1$

$e^{kT} = 3 \Rightarrow k = \frac{\ln 3}{2}$.

33 [Intermediate] • Target: Hypothesis test for a Poisson mean

Find the coefficient of x^2 in the binomial expansion of $(1 + x)^4$.

a. **6** ✓

b. 12

c. -6

d. $\frac{13}{2}$

Coefficient $= 42 = 122! = 6$.

34 [Intermediate] • Target: Further Poisson mean tests

The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. A new scoring algorithm is implemented, and the scores are transformed using the rule $Y = aX + b$, where X is the original score and Y is the new score. The new distribution of scores, Y, also follows a normal distribution with a mean of 120 and a standard deviation of 22.5. Calculate the value of a and the value of b.

a. **a = 1.5, b = -30** ✓

b. a = -1.5, b = 270

c. a = 1.5, b = 30

d. a = 0.67, b = 53

Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$. We are given $\mu_X = 100$, $\sigma_X = 15$, $\mu_Y = 120$, and $\sigma_Y = 22.5$. The linear transformation is given by $Y = aX + b$. We know that for a linear transformation $Y = aX + b$, the mean of Y is $\mu_Y = a\mu_X + b$ and the standard deviation of Y is $\sigma_Y = |a|\sigma_X$. From the standard deviation relationship: $\sigma_Y = |a|\sigma_X$ $22.5 = |a| \times 15$ $|a| = \frac{22.5}{15} = 1.5$ Now consider the mean relationship.

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35 [Intermediate] • Target: Hypothesis test for a binomial proportion

Line $l_1 : \mathbf{r} = (-3, 2, -2) + t(1, 3, 1)$ and line $l_2 : \mathbf{r} = (-7, 1, -5) + s(-3, 2, -2)$ intersect at one point. Find the y-coordinate of the point of intersection.

a. **-1** ✓

b. -4

c. -3

d. 0

The lines meet at $t = -1$, giving the point $(-4, -1, -3)$. y-coordinate: -1.

36 [Intermediate] • Target: Further binomial proportion tests

Find the coefficient of x^3 in the Maclaurin series of $\cos x$.

a. $-\frac{1}{6}$

b. $\frac{1}{6}$

c. $\frac{1}{3}$

d. **0** ✓

$\cos x$ has only even-power terms, so the coefficient of x^3 is 0.

37 [Intermediate] • Target: Type I and Type II errors, and significance level

Line $l_1 : \mathbf{r} = (-7, -11, 3) + t(-1, -3, -1)$ and line $l_2 : \mathbf{r} = (-1, -5, 3) + s(-1, 1, 1)$ intersect at one point. Find the x-coordinate of the point of intersection.

a. 6

b. -3

c. -2

d. **-4** ✓

The lines meet at $t = -3$, giving the point $(-4, -2, 6)$. x-coordinate: -4.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: testing a new manufacturing process • Cross-Lesson: 5.01 Integration

Describe the locus $|z - (2i)| = 1$ in the Argand diagram. Find its radius.

a. **1** ✓

b. 4

c. 2

d. 5

The locus $|z - z_0| = r$ is a circle centered at $z_0 = 2i$ with radius $r = 1$.

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39 [Challenging] • Target: The logic of hypothesis testing • Cross-Lesson: 6.01 Sequences & Series

A sequence satisfies $u_{n+1} = \frac{1}{2}u_n + 8$ with $u_1 = 5$. Find u_3.

a. $\frac{33}{4}$

b. $\frac{85}{4}$

c. 5

d. $\frac{53}{4}$ ✓

Applying the recurrence 2 times from $u_1 = 5$ gives $u_3 = \frac{53}{4}$.

40 [Challenging] • Target: Hypothesis test for a population mean (known σ) • Cross-Lesson: 7.01 Vectors

Differentiate $y = (x)^2$ and evaluate $\frac{dy}{dx}$ at $x = 0$.

a. 4

b. 5

c. 1

d. **0** ✓

$\frac{dy}{dx} = 2 \times 1 \times (x)^1$. At $x = 0$: $(x) = 0$, giving 0.

41 [Challenging] • Target: Two-tailed tests for a population mean • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A geometric sequence has first term $a = 1$ and common ratio $r = 2$. Find the sum of the first 5 terms.

a. **31** ✓

b. -31

c. 32

d. 16

$S_n = \frac{a(r^n - 1)}{r - 1} \Rightarrow S_5 = \frac{((2)^5 - 1)}{2 - 1} = 31$.

42 [Challenging] • Target: Hypothesis test for a Poisson mean • Cross-Lesson: 9.01 Complex Numbers

Given that $(x + 3)$ is a factor of $P(x) = 4x^3 - 3x^2 - x + m$, find the value of m.

a. **132** ✓

b. 129

c. 136

d. -132

By the Factor Theorem: $P(-3) = 0 \Rightarrow -132 + m = 0 \Rightarrow m = 132$.

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43 [Challenging] • Target: Further Poisson mean tests • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Line $l_1 : \mathbf{r} = (0, 7, -2) + t(-1, 1, -2)$ and line $l_2 : \mathbf{r} = (-1, 4, 6) + s(1, 1, -3)$ intersect at one point. Find the z-coordinate of the point of intersection.

a. 6

b. 4

c. **0** ✓

d. 1

The lines meet at $t = -1$, giving the point $(1, 6, 0)$. z-coordinate: 0.

44 [Challenging] • Target: Hypothesis test for a binomial proportion • Cross-Lesson: 10.01 Numerical Methods

Given $\tan\theta = \frac{5}{3}$ with θ in Quadrant III, state the sign of $\sec\theta$.

a. zero

b. **negative** ✓

c. positive

d. undefined

Quadrant III: $\cos\theta$ is negative, so $\sec\theta$ is negative.

45 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 11.01 Differential Equations

Given $\tan\theta = -\frac{5}{3}$ with θ in Quadrant IV, state the sign of $\sec\theta$.

a. negative

b. zero

c. **positive** ✓

d. undefined

Quadrant IV: $\cos\theta$ is positive, so $\sec\theta$ is positive.

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